Fixed Point Method

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1 Introduction

The Fixed-Point Iteration Method is an iterative numerical technique for solving nonlinear equations of the form

$$f(x) = 0.$$

Unlike the Bisection, False Position, and Secant methods, the Fixed-Point Iteration method first transforms the equation into the equivalent form

$$x = g(x),$$

where $g(x)$ is called the iteration function. Starting from an initial guess x_0, the method repeatedly computes

$$x_{n+1} = g(x_n)$$

until successive approximations become sufficiently close. A value x^* satisfying $g(x^*) = x^*$ is called a **fixed point** of the function $g(x)$. If the iteration converges, this fixed point is also a solution of the original equation $f(x) = 0$.

One important characteristic of the Fixed-Point Iteration method is that the convergence depends strongly on the choice of the iteration function $g(x)$. Different rearrangements of the same equation may lead to completely different convergence behaviors. Some choices converge rapidly, while others may converge slowly or even diverge.

2 Mathematical Formulation

2.1 Rearranging the Equation

The Fixed-Point Iteration Method differs from many other root-finding methods in that it does not operate directly on the equation $f(x) = 0$. Instead, the equation is first transformed into the equivalent form $x = g(x)$.

For example, consider the nonlinear equation

$$x^2 - 3x + 2 = 0.$$

It can be rewritten in several equivalent forms, such as $x = \frac{x^2+2}{3}$, $x = \sqrt{3x-2}$ or $x = 3 - \frac{2}{x}$. Each of these equations defines a different iteration function $g(x)$. Starting from the same initial approximation, one iteration function may converge rapidly to the solution, another may converge slowly, while a third may fail to converge altogether.

2.2 Convergent and Divergent Iterations

To illustrate the importance of choosing an appropriate iteration function, consider again the nonlinear equation $x^2 - 3x + 2 = 0$, whose roots are $x = 1, x = 2$.

First, choose the iteration function $x = \frac{x^2+2}{3}$, and let the initial approximation be $x_0 = 2.5$. The first few iterations are

$$x_1 = \frac{2.5^2 + 2}{3} = 2.75,$$

$$x_2 = \frac{2.75^2 + 2}{3} = 3.1875,$$

$$x_3 = \frac{3.1875^2 + 2}{3} = 4.0521, \ldots$$

The approximations move farther and farther away from the root instead of approaching it.

Now consider another rearrangement, $x = \sqrt{3x - 2}$, using the same initial approximation $x_0 = 2.5$. The iterations become

$$x_1 = \sqrt{5.5} = 2.3452,$$

$$x_2 = \sqrt{5.0356} = 2.2440,$$

$$x_3 = \sqrt{4.7320} = 2.1753, \ldots$$

In this case, the approximations move closer and closer to the root $x = 2$, indicating that the iteration converges.

3 Algorithm Flow

Starting from an initial approximation x_0, the Fixed-Point Iteration Method repeatedly evaluates the iteration function $x_{n+1} = g(x)$, to generate a sequence of approximations. After each iteration, the difference between two successive approximations is checked against a prescribed tolerance. If the difference is smaller than the tolerance, the current approximation is accepted as the solution. Otherwise, the newly computed approximation becomes the starting point for the next iteration. The process continues until the convergence criterion is satisfied or the maximum number of iterations is reached.

Algorithm 1 Fixed Point Method

Input: Continuous iteration function $g(x)$, initial approximation x_0
Output: Approximate fixed point x

```
while not converged do
    x_1 = g(x_0)
    if |x_1 - x_0| < ϵ then
        return x_1
    end if
    x_0 = x_1
end while
return x_1
```

4 Convergence Analysis

4.1 Limit

Let $f(x)$ be defined for all $x \neq a$ over an open interval containing a, and let L be a real number. Then

$$\lim_{x \to a} f(x) = L$$

if, for every $\epsilon > 0$, there exists a $\delta > 0$ such that if $0 < |x - a| < \delta$ then $|f(x) - L| < \epsilon$. This definition means that the function values $f(x)$ can be made arbitrarily close to the limit L by choosing x sufficiently close to a, but not equal to a.

Properties of Limits
If L, M, a are real numbers and

$$\lim_{x \to a} f(x) = L \text{ and } \lim_{x \to a} g(x) = M, \text{ then}$$

1. Sum Rule: $\lim_{x \to a}[f(x) + g(x)] = L + M$
2. Difference Rule: $\lim_{x \to a}[f(x) - g(x)] = L - M$
3. Product Rule: $\lim_{x \to a}[f(x)g(x)] = LM$
4. Quotient Rule: $\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{L}{M}$, where $M \neq 0$

4.2 Continuity

We noticed that the limit of a function as x approaches a can often be found simply by calculating the value of the function at a. Functions with this property are called continuous at a.

A function is **continuous at a number** a if

$$\lim_{x \to a} f(x) = f(a).$$

A function $f(x)$ is continuous at $x = a$ if the following three conditions are satisfied:
1. $f(a)$ is defined
2. $\lim_{x \to a} f(x)$ exist
3. $\lim_{x \to a} f(x) = f(a)$

4.3 Differentiability

A function is said to be differentiable at a point $x = a$ if its derivative exists at that point. The derivative of a function g(x) is defined by

$$g'(a) = \lim_{h \to 0} \frac{g(a + h) - g(a)}{h}.$$

If a function is differentiable at a point, then it is also continuous at that point. However, the converse is not necessarily true; a continuous function may fail to be differentiable.

4.4 Convergence

The convergence of the method depends on the properties of the iteration function $g(x)$, particularly its behavior near the fixed point.

Suppose the iteration $x_{n+1} = g(x_n)$ converges to a fixed point x^*. Since $x^* = g(x^*)$, consider a small error $e_n = x_n - x^*$. The error at the next iteration is $e_{n+1} = x_{n+1} - x^*$. Since the Fixed-Point Iteration Method computes $x_{n+1} = g(x_n)$, we obtain $e_{n+1} = g(x_n) - g(x^*)$.

Since $g(x)$ is differentiable, we can approximate $g(x)$ by the first-order Taylor expansion about the fixed point x^*:

$$g(x_n) = g(x^*) + g'(x^*)(x_n - x^*) + O(e_n^2).$$

Substituting this expansion into the error equation gives $e_{n+1} = g'(x^*)(x_n - x^*) + O(e_n^2)$. Since $e_n = x_n - x^*$, we obtain $e_{n+1} = g'(x^*)e_n + O(e_n^2)$. When the error is sufficiently small, the higher-order term can be neglected. Therefore,

$$e_{n+1} \approx g'(x^*)e_n.$$

Taking absolute values,

$$|e_{n+1}| \approx |g'(x^*)||e_n|.$$

If $|g'(x^*)| < 1$ then $|e_{n+1}| < |e_n|$, so the error decreases with each iteration and the sequence converges. On the other hand, if $|g'(x^*)| > 1$ then $|e_{n+1}| > |e_n|$, so the error increases and the iteration diverges. If $|g'(x^*)| = 1$ the convergence behavior is inconclusive. The iteration may converge slowly, oscillate, or diverge depending on the specific iteration function.

Therefore, the convergence of the Fixed-Point Iteration Method is determined by the magnitude of the derivative of the iteration function near the fixed point. A suitable iteration function should satisfy $|g'(x^*)| < 1$ to ensure convergence.

4.5 Guaranteed Convergence

In practice, the exact fixed point x^* is unknown, so the condition $|g'(x^*)| < 1$ cannot be verified directly. Instead, a stronger condition is checked on an interval $[a, b]$ that contains the fixed point. If the following conditions are satisfied:

1. $g([a, b]) \subseteq [a, b]$, and
2. $|g'(x)| \leq k < 1 \ \forall x \in [a, b]$,

then every iterate remains inside the interval, and the error decreases throughout the iteration.

4.6 Guaranteed Convergence - Example

Consider the iteration function $g(x) = \sqrt{x + 1}$ on the interval $[a, b]$.

1. Verify that the iteration function maps the interval into itself : $g([1, 2]) \subseteq [1, 2]$.
Evaluate the function at the endpoints:

$$g(1) \approx 1.4142,$$

$$g(2) \approx 1.7321.$$

$g(x)$ is increasing on the interval, it follows that $g([1, 2]) \subseteq [1, 2]$

2. Verify that $|g'(x)| \leq k < 1$
Differentiate the iteration function:

$$g'(x) = \frac{1}{2\sqrt{x+1}}$$

For every $x \in [1, 2]$,

$$\frac{1}{2\sqrt{3}} \leq g'(x) \leq \frac{1}{2\sqrt{2}}.$$

Therefore,

$$g'(x) \leq \frac{1}{2\sqrt{2}} \approx 0.3536 < 1.$$

Both convergence conditions are satisfied: $g([1, 2]) \subseteq [1, 2]$ and $|g'(x)| \leq 0.3536 < 1$. Hence, for any initial approximation $x_0 \in [1, 2]$, the Fixed-Point Iteration Method is guaranteed to converge to the unique fixed point $x^* = \frac{1 \pm \sqrt{5}}{2} \approx 1.61803$.

5 Numerical Example

To illustrate the Fixed Point Method, consider the nonlinear equation

$$f(x) = x^2 - x - 1.$$

The exact root is

$$x = \frac{1 \pm \sqrt{5}}{2}.$$

Rearranging the equation gives the iteration function

$$g(x) = \sqrt{x+1},$$

and let the initial approximation be $x_0 = 1$.

Applying the Fixed Point Method produces the following sequence of approximations.

Table 1: $g(x) = \sqrt{x+1}$

Iteration	Approximation	$g(x)$
1	1	1.41421
2	1.41421	1.55377
3	1.55377	1.59805
$\vdots$	$\vdots$	$\vdots$
17	1.61803	1.61803

After 17 iterations, the approximate root is 1.61803.